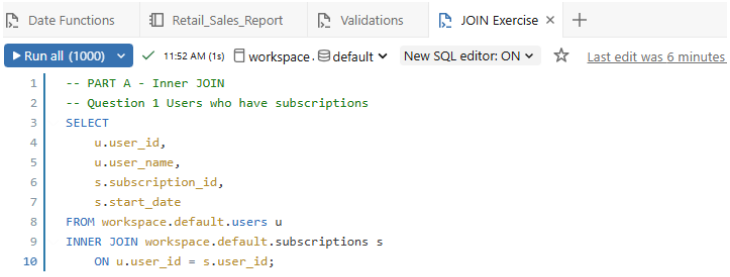
Objective

The objective of this exercise is to develop an understanding of how INNER JOIN, LEFT JOIN, and FULL OUTER JOIN are used to combine data from multiple tables. Also to strengthen practical SQL skills by writing queries that retrieve, match, and identify missing relationships between tables.

Question 1

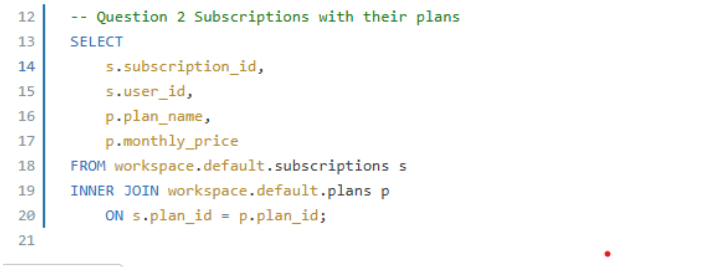


Add parameter

Table

	user_id	user_name	subscription_id	start_date
1	1	Nomvula	503	2026-03-10T00:00:00.000+00:...
2	2	David	502	2026-02-01T00:00:00.000+00:...
3	3	Anele	505	2026-04-05T00:00:00.000+00:...
4	1	Nomvula	501	2026-01-15T00:00:00.000+00:...

Question 2



Add parameter

Table

	subscription_id	user_id	plan_name	monthly_price
1	501	1	Basic	79
2	504	6	Standard	129
3	503	1	Premium	199
4	505	3	Family	249
5	502	2	Standard	129

Question 3

```sql
--Question 3 Viewing sessions with matching shows
SELECT
    v.session_id,
    v.user_id,
    sh.show_title,
    sh.genre,
    v.watch_minutes
FROM workspace.default.viewing_sessions v
INNER JOIN workspace.default.shows sh
    ON v.show_id = sh.show_id;

```

Add parameter

Table +

	session_id	user_id	show_title	genre	watch_minutes
1	904	7	Comedy Hour	Comedy	20
2	903	1	Crime Time	Drama	60
3	902	2	Tech Tales	Documentary	30
4	905	3	Wild Earth	Documentary	90
5	901	1	Comedy Hour	Comedy	45

Question 4

```sql
-- Question 4 Viewing sessions with matching users
SELECT
    u.user_name,
    u.country,
    v.session_id,
    v.show_id,
    v.watch_minutes
FROM workspace.default.users u
INNER JOIN workspace.default.viewing_sessions v
    ON u.user_id = v.user_id;
```

Add parameter

Table +

	user_name	country	session_id	show_id	watch_minutes
1	Nomvula	Johannesburg	903	702	60
2	David	Cape Town	902	703	30
3	Anele	Durban	905	705	90
4	Nomvula	Johannesburg	901	701	45

Question 5

```sql
--Question 5 Users, subscriptions and plans
SELECT
    u.user_name,
    u.country,
    p.plan_name,
    p.monthly_price,
    s.start_date
FROM users u
INNER JOIN workspace.default.subscriptions s
    ON u.user_id = s.user_id
INNER JOIN workspace.default.plans p
    ON s.plan_id = p.plan_id;
```

Add parameter

Table +

	user_name	country	plan_name	monthly_price	start_date
1	Nomvula	Johannesburg	Premium	199	2026-03-10T00:00:00.000+00:...
2	David	Cape Town	Standard	129	2026-02-01T00:00:00.000+00:...
3	Anele	Durban	Family	249	2026-04-05T00:00:00.000+00:...
4	Nomvula	Johannesburg	Basic	79	2026-01-15T00:00:00.000+00:...

Question 6

```sql
--PART B - LEFT JOIN
--Question 6 - All users and subscriptions
SELECT
    u.user_id,
    u.user_name,
    s.subscription_id,
    s.start_date
FROM workspace.default.users u
LEFT JOIN workspace.default.subscriptions s
    ON u.user_id = s.user_id;
```

Add parameter

Table

	user_id	user_name	subscription_id	start_date
1	1	Nomvula	503	2026-03-10T00:00:00.000+00:...
2	2	David	502	2026-02-01T00:00:00.000+00:...
3	3	Anele	505	2026-04-05T00:00:00.000+00:...
4	4	Kabelo	null	null
5	5	Lerato	null	null
6	1	Nomvula	501	2026-01-15T00:00:00.000+00:...

Question 7

```sql
--Question 7 All plans and subscriptions
SELECT
    p.plan_id,
    p.plan_name,
    s.subscription_id,
    s.user_id
FROM workspace.default.plans p
LEFT JOIN workspace.default.subscriptions s
    ON p.plan_id = s.plan_id;

--Question 8
```

Add parameter

Table

	plan_id	plan_name	subscription_id	user_id
1	10	Basic	501	1
2	11	Standard	504	6
3	12	Premium	503	1
4	13	Family	505	3
5	14	Mobile	null	null
6	11	Standard	502	2

Question 8

```sql
--Question 8 All shows and viewing sessions
SELECT
    sh.show_id,
    sh.show_title,
    v.session_id,
    v.watch_minutes
FROM workspace.default.shows sh
LEFT JOIN workspace.default.viewing_sessions v
    ON sh.show_id = v.show_id;
```

Add parameter

Table

	show_id	show_title	session_id	watch_minutes
1	701	Comedy Hour	904	20
2	702	Crime Time	903	60
3	703	Tech Tales	902	30
4	704	Cooking Lab	null	null
5	705	Wild Earth	905	90
6	701	Comedy Hour	901	45

Question 9

```sql
--Question 9 All viewing sessions with users
SELECT
    v.session_id,
    v.show_id,
    v.watch_minutes,
    u.user_id,
    u.user_name
FROM workspace.default.viewing_sessions v
LEFT JOIN workspace.default.users u
    ON v.user_id = u.user_id;

```

Add parameter

Table

	session_id	show_id	watch_minutes	user_id	user_name
1	901	701	45	1	Nomvula
2	902	703	30	2	David
3	903	702	60	1	Nomvula
4	904	701	20	null	null
5	905	705	90	3	Anele

Question 10

```sql
--Question 10 All users and their plans
SELECT
    u.user_name,
    u.country,
    p.plan_name,
    p.monthly_price
FROM workspace.default.users u
LEFT JOIN workspace.default.subscriptions s
    ON u.user_id = s.user_id
LEFT JOIN workspace.default.plans p
    ON s.plan_id = p.plan_id;

```

Add parameter

Table

	user_name	country	plan_name	monthly_price
1	Nomvula	Johannesburg	Premium	199
2	David	Cape Town	Standard	129
3	Anele	Durban	Family	249
4	Kabelo	Pretoria	null	null
5	Lerato	Port Elizabeth	null	null
6	Nomvula	Johannesburg	Basic	79

Question 11

```sql
--PART C FULL OUTER JOIN
-- Question 11 FULL OUTER JOIN users and subscriptions
SELECT
    COALESCE(u.user_id, s.user_id) AS user_id,
    u.user_name,
    s.subscription_id,
    s.start_date
FROM workspace.default.users u
FULL OUTER JOIN workspace.default.subscriptions s
    ON u.user_id = s.user_id;

```

Add parameter

Table

	user_id	user_name	subscription_id	start_date
1	3	Anele	505	2026-04-05T00:00:00.000+00:...
2	4	Kabelo	null	null
3	2	David	502	2026-02-01T00:00:00.000+00:...
4	5	Lerato	null	null
5	1	Nomvula	503	2026-03-10T00:00:00.000+00:...
6	1	Nomvula	501	2026-01-15T00:00:00.000+00:...
7	6	null	504	2026-03-20T00:00:00.000+00:...

Question 12

```sql
-- Question 12 FULL OUTER JOIN plans and subscriptions
SELECT
    COALESCE(p.plan_id, s.plan_id) AS plan_id,
    p.plan_name,
    s.subscription_id,
    s.user_id
FROM workspace.default.plans p
FULL OUTER JOIN workspace.default.subscriptions s
    ON p.plan_id = s.plan_id;

```

Add parameter

Table

	plan_id	plan_name	subscription_id	user_id
1	11	Standard	504	6
2	14	Mobile	null	null
3	13	Family	505	3
4	12	Premium	503	1
5	10	Basic	501	1
6	11	Standard	502	2

Question 13

```sql
-- Question 13 FULL OUTER JOIN shows and viewing sessions
SELECT
    COALESCE(sh.show_id, v.show_id) AS show_id,
    sh.show_title,
    v.session_id,
    v.watch_minutes
FROM workspace.default.shows sh
FULL OUTER JOIN workspace.default.viewing_sessions v
    ON sh.show_id = v.show_id;


```

Add parameter

Table

	show_id	show_title	session_id	watch_minutes
1	704	Cooking Lab	null	null
2	705	Wild Earth	905	90
3	701	Comedy Hour	904	20
4	703	Tech Tales	902	30
5	702	Crime Time	903	60
6	701	Comedy Hour	901	45

Question 14

```sql
--Question 14 FULL OUTER JOIN users and sessions
SELECT
    COALESCE(u.user_id, v.user_id) AS user_id,
    u.user_name,
    v.session_id,
    v.show_id,
    v.watch_minutes
FROM workspace.default.users u
FULL OUTER JOIN workspace.default.viewing_sessions v
    ON u.user_id = v.user_id;


```

Add parameter

Table

	user_id	user_name	session_id	show_id	watch_minutes
1	3	Anele	905	705	90
2	2	David	902	703	30
3	7	null	904	701	20
4	1	Nomvula	901	701	45
5	1	Nomvula	903	702	60
6	4	Kabelo	null	null	null
7	5	Lerato	null	null	null

Question 15

```sql
-- Question 15 Users, subscriptions and plans using FULL OUTER JOIN
SELECT
    COALESCE(u.user_id, s.user_id) AS user_id,
    u.user_name,
    s.subscription_id,
    COALESCE(p.plan_id, s.plan_id) AS plan_id,
    p.plan_name
FROM workspace.default.users u
FULL OUTER JOIN workspace.default.subscriptions s
    ON u.user_id = s.user_id
FULL OUTER JOIN workspace.default.plans p
    ON s.plan_id = p.plan_id;
```

Add parameter

Table +

	user_id	user_name	subscription_id	plan_id	plan_name
1	2	David	502	11	Standard
2	6	null	504	11	Standard
3	3	Anele	505	13	Family
4	1	Nomvula	503	12	Premium
5	4	Kabelo	null	null	null
6	5	Lerato	null	null	null
7	1	Nomvula	501	10	Basic
8	null	null	null	14	Mobile

Bonus Question 1

```sql
-- BONUS QUESTION
--BONUS Question 1: Which users have not subscribed to any plan?
SELECT
    u.user_id,
    u.user_name,
    u.country
FROM workspace.default.users u
LEFT JOIN workspace.default.subscriptions s
    ON u.user_id = s.user_id
WHERE s.subscription_id IS NULL;
```

Add parameter

Table +

	user_id	user_name	country
1	4	Kabelo	Pretoria
2	5	Lerato	Port Elizabeth

Bonus Question 2

```sql
--BONUS Question 2: Which subscriptions reference users that do not exist?
SELECT
    s.subscription_id,
    s.user_id,
    s.plan_id,
    s.start_date
FROM workspace.default.subscriptions s
LEFT JOIN workspace.default.users u
    ON s.user_id = u.user_id
WHERE u.user_id IS NULL;
```

Add parameter

Table +

	subscription_id	user_id	plan_id	start_date
1	504	6	11	2026-03-20T00:00:00.000+00:...

Bonus Question 3

```sql
--BONUS Question 3: Which shows have never been watched?
SELECT
    sh.show_id,
    sh.show_title,
    sh.genre
FROM workspace.default.shows sh
LEFT JOIN workspace.default.viewing_sessions v
    ON sh.show_id = v.show_id
WHERE v.session_id IS NULL;
```

Add parameter

Table

	show_id	show_title	genre
1	704	Cooking Lab	Lifestyle

Bonus Question 4

```sql
--BONUS Question 4: Which viewing sessions reference shows that do not exist?
SELECT
    v.session_id,
    v.user_id,
    v.show_id,
    v.watch_minutes
FROM workspace.default.viewing_sessions v
LEFT JOIN workspace.default.shows sh
    ON v.show_id = sh.show_id
WHERE sh.show_id IS NULL;
```

Add parameter

Table

session_id	user_id	show_id	watch_minutes

No rows returned

Bonus Question 5

```sql
--BONUS Question 5: Which plans have no subscribers?
SELECT
    p.plan_id,
    p.plan_name,
    p.monthly_price
FROM workspace.default.plans p
LEFT JOIN workspace.default.subscriptions s
    ON p.plan_id = s.plan_id
WHERE s.subscription_id IS NULL;
```

Add parameter

Table

	plan_id	plan_name	monthly_price
1	14	Mobile	59